

Chapter 10

From Information Geometry to Information Gravity

The Emergence of Gravitational Structure from Informational Geometry

10.1 Introduction

The Theory of Entropicity is built upon a simple but profound observation.

Modern physics has established two independent facts.

First, information possesses geometry.

Second, gravity is geometry.

These facts arise from entirely different areas of science.

The first emerges from information theory, statistical inference, and quantum theory.

The second emerges from General Relativity.

The central insight of the Theory of Entropicity is that these two facts may not be independent.

If information possesses geometry, and if geometry manifests physically as gravity, then informational geometry itself may possess a gravitational character.

This observation forms one of the deepest conceptual foundations of ToE.

The present chapter develops this idea rigorously.

10.2 The Foundational Syllogism

The conceptual core of the Theory of Entropicity may be expressed as the following logical sequence.

Premise I

Information possesses geometry.

Premise II

Geometry manifests physically as gravity.

Conclusion

Information geometry possesses gravitational structure.

Symbolically,

Information $\longrightarrow$ Geometry,

Geometry $\longrightarrow$ Gravity,

therefore,

Information Geometry $\longrightarrow$ Information Gravity.

This conclusion is not presented as a metaphor.

Within ToE it is proposed as a physical principle.

10.3 Information Possesses Geometry

The first premise is already established within modern mathematics.

The Fisher–Rao metric provides a geometry for classical probability distributions.

For a statistical manifold,

$$g_{ij}^{(FR)} = \int p(x|\theta) \partial_i \ln p \partial_j \ln p dx.$$

This metric measures distinguishability between probability distributions.

Distances correspond to informational separation.

Similarly, the Fubini–Study metric provides geometry for quantum states:

$$ds_{FS}^2 = 4 \left(\langle d\psi | d\psi \rangle - |\langle \psi | d\psi \rangle|^2 \right).$$

These structures are genuine geometries.

They possess:

- metrics,
- geodesics,
- curvature,
- connections,
- volumes.

Consequently information is not merely abstract.

Information occupies a geometrical manifold.

10.4 Geometry as a Physical Entity

The second premise emerges from General Relativity.

Einstein's theory established that gravity is not a force in the Newtonian sense.

Instead,

$$\text{Gravity} = \text{Spacetime Geometry.}$$

The Einstein Field Equations,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

demonstrate that curvature possesses direct physical consequences.

Matter curves spacetime.

Spacetime curvature governs motion.

Geometry therefore becomes a dynamical physical entity.

This observation fundamentally altered our understanding of nature.

10.5 Combining the Two Premises

The crucial question now arises.

If information possesses geometry, and geometry possesses physical significance, what prevents informational geometry from possessing physical significance?

From the perspective of ToE, nothing prevents such an interpretation.

Indeed, the repeated appearance of information and entropy in modern gravitational physics suggests precisely this possibility.

The theory therefore proposes that informational geometry is not merely descriptive.

It is dynamical.

10.6 Definition of Information Gravity

Formal Definition

Information Gravity is defined as the gravitational structure associated with informational geometry.

It represents the curvature generated within informational manifolds.

Symbolically,

$$\mathcal{G}_{IG} = \text{Curvature of Information Space.}$$

The concept applies to both:

$$FR$$

and

$$FS$$

sectors.

Thus information gravity exists in both classical and quantum informational domains.

10.7 Fisher–Rao Information Gravity

Consider the Fisher–Rao manifold.

Its metric

$$g_{ij}^{(FR)}$$

possesses curvature.

From this metric one may construct:

$$\begin{aligned} R^{(FR)\rho}_{\sigma\mu\nu}, \\ R^{(FR)}_{\mu\nu}, \\ R^{(FR)}. \end{aligned}$$

These are directly analogous to the curvature quantities appearing in General Relativity.

ToE interprets these objects as manifestations of classical information gravity.

Regions of high informational curvature correspond to regions of strong informational organization.

10.8 Fubini–Study Information Gravity

The same reasoning applies to the quantum sector.

The Fubini–Study metric generates:

$$\begin{aligned} R^{(FS)\rho}_{\sigma\mu\nu}, \\ R^{(FS)}_{\mu\nu}, \\ R^{(FS)}. \end{aligned}$$

These curvature quantities characterize the geometry of quantum state space.

Within ToE they are interpreted as manifestations of quantum information gravity.

Quantum informational structure therefore acquires geometric and potentially gravitational significance.

10.9 Information Curvature

The central object underlying information gravity is informational curvature.

For a sector J ,

$$R^{(J)}$$

measures the curvature of informational space.

In conventional information geometry this quantity is often interpreted statistically.

ToE proposes a stronger interpretation.

Informational curvature represents a physical tendency toward organization and constraint.

The greater the informational curvature, the stronger the informational gravitational effect.

10.10 Entropy as the Source of Information Gravity

The Theory of Entropicity does not stop at information geometry.

It asks a deeper question.

What generates informational geometry itself?

The proposed answer is entropy.

Entropy organizes information.

Information acquires geometry.

Geometry acquires curvature.

Curvature manifests as gravity.

The chain becomes

$$S \longrightarrow I \longrightarrow G_I \longrightarrow \mathcal{G}_{IG}.$$

This sequence represents the entropic origin of information gravity.

10.11 Informational Geodesics

In General Relativity, particles follow geodesics determined by spacetime geometry.

Similarly, informational states follow geodesics within informational manifolds.

The geodesic equation is

$$\frac{d^2 x^\rho}{d\lambda^2} + \Gamma_{\mu\nu}^\rho \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = 0.$$

Within the Fisher–Rao and Fubini–Study sectors, this equation describes optimal informational evolution.

ToE interprets such trajectories as informational analogues of gravitational motion.

10.12 Information Gravity and Entropic Flow

Entropy gradients influence informational organization.

Informational organization influences informational curvature.

Therefore entropy gradients generate informational gravity.

Symbolically,

$$\nabla S / 0 =$$

implies

$$\nabla I / 0, =$$

which implies

$$R^{(I)} / 0. =$$

Consequently informational gravitational fields emerge.

10.13 The Information Gravity Tensor

To formalize this concept, ToE introduces the informational Einstein tensor:

$$G_{\mu\nu}^{(I)} = R_{\mu\nu}^{(I)} - \frac{1}{2} g_{\mu\nu}^{(I)} R^{(I)}.$$

This tensor measures informational gravitational structure.

For HMAS,

$$G_{\mu\nu}^{(I)} = G_{\mu\nu}^{(FR)} + G_{\mu\nu}^{(FS)}.$$

It represents the total information-gravity contribution.

10.14 Information Gravity and Spacetime Gravity

The next conceptual step is crucial.

If informational geometry possesses gravitational structure, then spacetime gravity may emerge from information gravity.

The proposed hierarchy becomes

$$G^{(FR)} + G^{(FS)} \longrightarrow G^{(L)}.$$

The Lorentzian sector is therefore interpreted as a macroscopic manifestation of deeper informational gravitational structures.

This idea lies at the heart of ToE.

10.15 Emergent Einstein Gravity

The informational sectors influence the Lorentzian sector through the Local Obidi Action.

The resulting Obidi Field Equations connect informational curvature and spacetime curvature.

In the low-entropy-gradient regime,

$$G^{(FR)} \rightarrow 0,$$

$$G^{(FS)} \rightarrow 0.$$

Only the Lorentzian contribution remains.

The theory therefore reduces to

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

Einstein gravity emerges naturally.

10.16 Comparison with Existing Approaches

Many modern approaches recognize a connection between information and gravity.

Jacobson derives Einstein gravity from thermodynamics.

Verlinde derives gravity from entropy gradients.

Bianconi derives modified gravitational equations from information-theoretic constructions.

The Theory of Entropicity differs in a significant way.

Rather than treating information as a tool for deriving gravity, ToE proposes that information geometry itself possesses intrinsic gravitational structure.

This distinction motivates the concept of Information Gravity.

10.17 The Information Gravity Correspondence Principle

The preceding discussion suggests a new principle.

Information Gravity Correspondence Principle

Whenever information possesses nontrivial geometry, it possesses an associated gravitational structure.

Symbolically,

$$R^{(I)} \neq 0 \implies \mathcal{G}_{IG} \neq 0. =$$

This principle provides a bridge between information geometry and gravitational physics.

10.18 Conceptual Summary

The argument developed throughout this chapter may be summarized as follows.

1. Information possesses geometry.
2. Geometry possesses gravitational significance.
3. Therefore informational geometry possesses gravitational significance.
4. Entropy generates informational organization.
5. Informational organization generates informational geometry.
6. Informational geometry generates information gravity.
7. Information gravity generates spacetime gravity.

Thus

$$S \longrightarrow I \longrightarrow G_I \longrightarrow \mathcal{G}_{IG} \longrightarrow G_L.$$

This chain represents one of the foundational conceptual structures of the Theory of Entropicity.

10.19 Transition to Entropic Gravity

Information Gravity establishes the connection between informational geometry and gravitational structure.

A deeper question remains.

What generates information gravity itself?

According to ToE, the answer is entropy.

The next chapter develops the concept of **Entropic Gravity** within the Theory of Entropicity and demonstrates how entropy gradients generate informational curvature, informational gravity, and ultimately spacetime gravitation.

There the theory moves from informational geometry to its deepest proposed foundation: entropy itself.